MATH 3310 HOMEWORK ASSIGNMENT 11

DUE ON TUESDAY 22 NOVEMBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

(1) For each of the following relations on $\{a, b, c, d, e\}$, decide which, if any, of the properties *reflexive*, *symmetric*, or *transitive* it has.
   (a) $\{(a, a), (b, b), (c, c)\}$.
   (b) $\{(a, a), (b, b), (c, c), (c, d), (d, c)\}$.
   (c) $\{(a, b), (b, c), (c, d), (d, e), (d, a)\}$.
   (d) $\{(a, b), (b, a), (c, b), (b, c), (a, c), (c, a)\}$.
   (e) $\{(a, a), (b, b), (c, c), (c, d), (d, d), (e, e), (a, b), (e, a)\}$.
   (f) $\{(a, a), (b, b), (c, c), (c, d), (d, d), (e, e), (e, a)\}$.

(2) Let A be a set. Prove that the relation $\approx$ on $\mathcal{P}(A)$ given by

$$X \approx Y \quad \text{if} \quad |X| = |Y|$$

is an equivalence relation.

(3) For every natural number n set

$$a_n = \sum_{i=1}^{n} \frac{1}{(2i-1)(2i+1)} \, .$$

   (a) Compute the numbers a_1, a_2, a_3, and a_4.
   (b) Conjecture a closed form expression for a_n.
   (c) Prove the formula conjectured in part (b).

(4) Consider the relation F on $\mathbb{R} \times \mathbb{R}$ given by

$$F = \{(x, e^x) \mid x \in \mathbb{R}\} \, .$$

   (a) Decide if F is an equivalence relation.
   (b) Determine the inverse relation F^{-1}.

(5) Let R be the relation on $\mathbb{N}$ given by

$$a \; R \; b \quad \text{if} \quad a|5b \text{ or } b|5a.$$

Decide if it is an equivalence relation.